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Impact of sex on outcomes in patients treated with drug-coated balloons versus drug-eluting stents for de novo coronary artery lesions: Insights from the REC-CAGEFREE I trial

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Running title: Sex impact on DCB versus DES outcomes

Abstract

Background

The anatomical and pathophysiological characteristics of coronary artery disease vary between the sexes. This study investigated the impact of sex on outcomes in patients with *de novo* coronary artery lesions treated with drug-coated balloons (DCB) or drug-eluting stents (DES).

Methods

REC-CAGEFREE I was an investigator-initiated, non-inferiority trial conducted at 43 sites in China from Feb 5, 2021, to May 1, 2022, which randomized 2,272 patients for treating *de novo* coronary lesions, regardless of vessel diameter. After successful lesion pre-dilatation, eligible patients were randomized (1:1) to either DCB angioplasty with the option of rescue stenting or intended DES deployment. In this prespecified subgroup analysis, patients were analyzed by sex based on their medical records. The primary endpoint was device-oriented composite endpoint (DoCE), including cardiovascular death, target-vessel myocardial infarction, and clinically and physiologically indicated target lesion revascularization at 2 years. Between-group differences were compared by Cox proportional-hazards models, and imbalances in baseline characteristics were adjusted with inverse probability of treatment weighting (IPTW). The analyses were conducted in the intention-to-treat population.

Results

A total of 2,272 participants underwent randomization, of which 698 (30.7%) were female and 1,574 (69.3%) were male. At 2 years, no statistically significant differences in the incidence of DoCE were observed between sexes (36 [5.2%] for females and 74 [4.7%] for males, HR_{IPTW}:1.04, 95%CI:0.67 to 1.61, P=0.877). Compared with DES, DCB was associated with a numerically higher risk of DoCE in females (6.3% versus 3.9%, HR_{IPTW}:1.55, 95%CI:0.78 to 3.11, P=0.210) and a statistically significant higher risk in males (6.4% versus 3.1%, HR_{IPTW}:2.28, 95%CI:1.40 to 3.70, P=0.001), respectively, with no significant sex-by-treatment (DCB/DES) interaction observed ($P_{interaction}$=0.575). The prognosis of DCB and DES differed significantly between small vessel disease (SVD) and non-SVD among females ($P_{\text{interaction}}$=0.007), but not among males ($P_{\text{interaction}}$=0.408).

Conclusions

For patients with *de novo*, non-complex coronary artery disease, DCB was associated with a significantly higher risk of 2-year DoCE compared with DES in males, whereas a consistent but non-significant trend was observed in females.

Trial registration ClinicalTrials.gov identifier: NCT04561739

Keywords Sex, Drug-coated balloon, Drug-eluting stent, De novo coronary artery lesion, Small vessel disease

Background

Newer-generation drug-eluting stents (DES) remain the mainstay of percutaneous coronary intervention (PCI) in contemporary practice (1). However, the polymer coating of DES can lead to delayed endothelialization, which increases the risk of late thrombosis, whilst the metal scaffold can trigger an inflammatory response, leading to in-stent restenosis (2). These persisting problems with DES motivated the development of non-stent-based strategies, for example, drug-coated balloons (DCB), which allow the rapid delivery of antiproliferative drugs directly into the arterial wall, eliminating the need for a permanent scaffold. Clinical studies with DCB have shown promising results in treating in-stent restenosis (1) and small vessel disease (SVD) (3). We recently compared DCB angioplasty with rescue stenting to intended DES deployment for the treatment of *de novo* coronary artery disease. The results showed that DES was associated with a significantly lower risk of the device-oriented composite endpoint (DoCE) compared with DCB (4).

Unlike permanent metallic stents, the clinical efficacy of DCB relies not only on adequate lesion preparation but on achieving sustained lumen patency following angioplasty, which is primarily influenced by two key factors: the degree of early elastic recoil and the occurrence of flow-limiting dissections (5). Notably, sex-specific anatomical variations in coronary vasculature, including luminal diameters, thickness of arterial walls (6), and the susceptibility to dissection (7), may disproportionately affect outcomes after DCB treatment. In addition, emerging evidence suggests the need to consider other factors, such as hormonal status, microvascular function, and sex-specific differences in plaque composition between female and male coronary lesions (8). Whether these factors contribute to sex-based differences in prognosis following DCB treatment remains unclear. To address this knowledge gap, this study compares sex differences in clinical outcomes between DCB and DES therapies for *de novo*, non-complex coronary lesions.

Methods

Study design and population

This is a prespecified subgroup analysis of the REC-CAGEFREE I study. The REC-CAGEFREE I trial was an investigator-initiated, prospective, randomized, non-inferiority trial conducted at 43 participating sites in China from Feb 5, 2021, to May 1, 2022. The trial rationale, design, and main findings have been reported previously (4, 9). The REC-CAGEFREE I study was approved by the institutional review board at each participating institution. Written informed consent was obtained from all patients. The study complied with the Declaration of Helsinki and Good Clinical Practice. An independent data and safety monitoring committee oversaw the safety of all patients.

Patients eligible for enrolment were those requiring PCI due to acute or chronic coronary syndrome with *de novo*, non-complex target lesions, regardless of treated vessel diameter. Key exclusion criteria were patients under 18 years of age, cardiogenic shock, and in-stent restenosis requiring revascularization. The full inclusion and exclusion criteria are in the Additional file (Table S1). In the current study, data on patient sex were collected from medical records. To facilitate cross-study comparisons, we pre-specified the definition of a small vessel in the REC-CAGEFREE I study as a vessel requiring a device diameter < 3.0 mm, in accordance with the cutoff used in the BASKET SMALL 2 trial (10).

PCI procedure and concomitant medications

For patients randomized to the DCB group, procedures were required to adhere to the recommendations of the German Consensus Group on DCB interventions and the Third Report of the International DCB Consensus Group (2, 11). Patients randomized to DCB were treated with the paclitaxel-coated balloon (Swide, Shenqi Medical, Shanghai, China) (12). Following DCB angioplasty, rescue stenting was mandated in cases of (a) NHLBI Type D, E, or F dissection, (b) TIMI flow ≤ 2, or (c) residual stenosis > 30% by visual assessment. Patients randomized to DES were treated with a second-generation thin-strut sirolimus-eluting stent (Firebird 2, MicroPort, Shanghai, China) (13) per established DES guidelines (1).

Randomization was performed only after successful lesion preparation, as recommended by the current international consensus (2, 11).

All study participants received antithrombotic treatment following established guidelines (11). Each participant was prescribed dual-antiplatelet therapy with aspirin (100 mg once daily) and clopidogrel (75 mg once daily) or aspirin and ticagrelor (90 mg twice daily) for at least 1 month after PCI. Thereafter, patients were required to receive either aspirin (100 mg once daily) or clopidogrel (75 mg once daily) monotherapy indefinitely. Physicians had the discretion to prescribe additional medical treatments; however, it was strongly recommended that they adhere to guideline-directed medical therapy tailored to the patient's specific condition.

Follow-up

Follow-up visits were scheduled at 1 (± 14 days), 3, 6, 12, 18, and 24 (± 30 days) months post-randomization, preferably conducted on-site. Patients unable or unwilling to attend the outpatient clinic could have their visit replaced by a telephone call, except for the 1-, 12-, and 24-month visits. Any potential clinical endpoint identified during a telephone call required subsequent verification through hospital records. To facilitate the acquisition of outcomes and adherence to the prescribed medications, we developed a mobile application that functions through the WeChat platform. All participants were contacted monthly through this application and received a questionnaire to evaluate their health status and adherence. Any potential event reported via this platform would be promptly judged by the investigators and, if necessary, would lead to a clinic visit for further investigation.

Endpoints

The primary endpoint of this analysis was a device-oriented composite endpoint (DoCE), defined as the composite of cardiovascular death, target vessel myocardial infarction (TV-MI), and clinically and physiologically indicated target lesion revascularization (CPI-TLR) assessed at 2 years. Target lesion revascularization was considered clinically and physiologically indicated if associated with wire-based or angiography-derived fractional flow reserve of 0.80 or less, or diameter stenosis of 70% or more by quantitative coronary angiography (14). The secondary endpoints were rates of the individual components of the DoCE; patient-oriented composite endpoint (PoCE; the composite of all-cause death, any stroke, any myocardial infarction, and any revascularization) and its components; target vessel failure (TVF), defined as vessel-related cardiovascular death, TV-MI, and clinically and physiologically indicated target vessel revascularization (CPI-TVR); Bleeding Academic Research Consortium (BARC)-defined type 3 or 5 bleeding; BARC-defined type 2, 3, or 5 bleeding events and its components; net adverse clinical events (defined as PoCE and BARC-defined 3 or 5 bleeding events); and definite or probable device or vessel thrombosis.

Statistical analysis

Continuous variables, presented as mean (standard deviation, SD), were compared using the Student's t-test. Categorical data, summarized as absolute numbers and proportions (%), were compared using the Chi-square test. Fisher's exact test was employed when the expected frequency in any cell of the contingency table was less than 5. To assess the impact of sex on treatment effect (DCB vs. DES), the cumulative incidence of clinical events was calculated using the Kaplan-Meier method. Hazard ratios (HRs) and 95% confidence intervals (CIs) were generated with Cox proportional-hazards models. The proportional hazard assumptions for Cox models were tested using Schoenfeld residuals (Additional file: Table S2). Multivariable Cox regression and inverse probability of treatment weighting (IPTW) were performed to adjust for potential confounding bias. The variables included in these adjustments are listed in the Additional file (Methods). Given the exploratory nature of this subgroup analysis, which was aimed at generating hypotheses, no formal correction for multiple testing was applied (15). All tests were two-sided, and a p-value of <0.05 was considered statistically significant. All analyses were conducted with the statistical software package R version 4.4.1 (R Foundation for Statistical Computing, Vienna, Austria).

Results

Between February 5, 2021, and May 1, 2022, a total of 2272 patients were randomly assigned to treatment. Of these participants, 698 (30.7%) were female. Among female and male patients, 35 (9.6%) and 71 (9.2%), respectively, allocated to the DCB group received rescue DES following suboptimal angioplasty (Fig. 1). Females were older and had a higher prevalence of hypertension, diabetes mellitus, hyperlipidemia, a higher left ventricular ejection fraction, and were more likely to present with unstable angina and chronic coronary syndrome. Males were more likely to be smokers and had a higher prevalence of previous myocardial infarction, chronic obstructive pulmonary disease, and peripheral vascular disease, and were more likely to present with ST-elevation and non-ST-elevation myocardial infarction. A total of 780 and 1749 lesions were treated in females and males, respectively. There were no significant differences in lesion location between the sexes. The use of cutting or scoring balloons was more common in males, who also had fewer lesions in SVD compared to females (45.3% vs. 55.5%; Table 1). The distribution of missing data is presented in the Additional file (Table S3).

Female patients randomized to the DCB group had a higher percentage of SVD compared with the DES group (60.0% vs. 50.4%, *P*=0.009). Male patients randomized to the DCB group were less likely to have diabetes and had a higher percentage of bifurcation disease than those randomized to the DES group. The total number and total length of devices per patient were greater in the DCB versus the DES groups, as was the likelihood of using non-compliant, cutting, or scoring balloons (Table 1). Baseline characteristics of female and male patients who underwent rescue DES in the DCB group are presented in the Additional file (Table S4). Rescue stent diameter was significantly smaller in female patients than in males. No significant differences in the indications for rescue stenting were observed between sexes. Nevertheless, a trend emerged whereby more males than females underwent rescue stenting due to residual stenosis, whereas a higher proportion of females required it for coronary dissection.

Two-year clinical follow-up (median 734 days) was available for 696 females (99.7%) and 1562 males (99.2%); two patients who withdrew consent and twelve who were lost to follow-up were censored at their last contact. At two years, the primary endpoint occurred in 36 of 698 females and 74 of 1574 males (5.2% vs. 4.7%, HR_{IPTW}, 1.04, 95% CI, 0.67 to 1.61, *P*=0.877); the incidence of PoCE, cardiovascular death, stroke, MI, TVF, CPI-TVR, net adverse clinical events, and BARC-defined type 3 or 5 bleeding was numerically similar between the sexes (Fig. 2, Additional file: Fig. S1). There was no significant difference in the rate of DoCE in female and male patients allocated to the DCB group who received rescue DES (*P*=0.672; Additional file: Fig. S2).

Compared with DES, DCB was associated with a numerically higher risk of DoCE in females (6.3% [23/364] vs. 3.9% [13/334], HR_{IPTW}:1.55, 95%CI:0.78 to 3.11, *P*=0.210) and a statistically significant higher risk in males (6.4% [49/769] vs. 3.1% [25/805], HR_{IPTW}:2.28, 95%CI:1.40 to 3.70, *P*=0.001), respectively, with no significant sex-by-treatment (DCB/DES) interaction ($P_{interaction}$=0.575; Fig. 3, Fig. 4). Compared with DES, the DCB strategy was associated with a higher risk of PoCE, target lesion revascularization, and target vessel revascularization in both males and females (Fig. 3, Additional file: Fig. S3). Both IPTW and multivariable Cox regression analyses revealed comparable rates of net adverse clinical events, death, stroke, and MI between DCB/DES strategies across sex subgroups (Additional file: Fig. S4; Fig. S5).

In males, the effect of DCB versus DES strategy on the primary endpoint did not differ statistically in both SVD (5.8% vs. 3.6%, HR_{IPTW} 1.80, 95% CI: 0.87 to 3.76, *P*=0.115) and non-SVD groups (6.9% vs. 2.8%, HR_{IPTW} 2.70, 95% CI: 1.40 to 5.23, *P*=0.003) ($P_{interaction}$=0.408). However, in females, a significant treatment-by-vessel-size interaction was observed ($P_{interaction}$=0.007). Compared to DES, the rate of DoCE was numerically lower with DCB in patients with SVD (4.0% vs. 6.1%, HR_{IPTW} 0.69, 95% CI: 0.26 to 1.85, *P*=0.465), and significantly higher in those without SVD (9.2% vs. 1.8%, HR_{IPTW} 5.42, 95% CI: 1.53 to 19.24, *P*=0.009) (Fig. 5).

Discussion

The main findings can be summarized as follows:

1) Female patients had different risk profiles compared with males;
2) At 2 years post-PCI, female and male patients demonstrated numerically similar rates of DoCE and other adverse clinical events;
3) No significant sex-by-treatment (DCB/DES) interaction was found. In male patients with *de novo* coronary lesions, the DCB strategy was associated with a significantly higher risk of 2-year DoCE as compared with DES; in female patients, the DoCE rate in the DCB group was numerically higher.
4) Among female participants, the prognosis of DCB and DES differed significantly between SVD and non-SVD. While such an effect was not observed in male participants.

In the current study, we found that male patients exhibit distinct cardiac risk factors from females, such as lower LVEF% and higher smoking rates. However, female patients were older than males, which is consistent with previous research conducted in Asian (16, 17) and Western populations (18, 19). Nevertheless, we did not observe sex-related differences in PCI outcomes, which aligns with previous meta-analyses (20, 21) and subgroup analyses from large-scale randomized studies (22-25). Studies from the plain old balloon angioplasty (POBA) era have identified female sex as an independent predictor of adverse outcomes after PCI (26); however, previous investigations reported comparable outcomes between female and male patients following DES implantation (21, 23, 27). Advancements in interventional techniques and evidence-based risk factor management may have mitigated this historical disparity. Notably, data regarding sex related outcomes after DCB treatment remain limited. A retrospective cohort study with DCB treatment documented greater late lumen loss in female patients at 6-month follow-up. Nevertheless, this anatomical difference did not translate into differences in target vessel failure (28). In our study, no significant differences in DoCE or target vessel failure were observed between females and males in either the DCB or DES groups. This observation contributes novel insights to the limited evidence of sex-specific outcomes after DCB/DES interventions.

The BASKET-SMALL II represents the largest RCT conducted to date comparing DCB and DES in *de novo* SVD only (10). The findings indicated that at three years, the major adverse cardiovascular events (MACE) rate for males was 15% in the DES cohort compared to 16% in the DCB cohort, while for females, the rates were 16% and 10%, respectively (p=0.38) (24). Although this difference was not statistically significant, there appeared to be a trend suggesting a reduced MACE rate in females with SVD who were treated with DCB. Our study, which involved a larger sample size, showed a significant interaction between SVD and treatment modality (DCB/DES) in females, a pattern not observed in males. While this interaction necessitates further validation, it hints at a potential biological rationale. Females typically have smaller coronary artery diameters and thinner vessel walls (6), which may predispose small vessels to stent-induced over-expansion damage and subsequent inflammation, exacerbating neointimal hyperplasia and restenosis (19, 29, 30). In contrast, DCB could mitigate these injuries through its "leave nothing behind" approach. Additionally, the drug penetration of DCB may benefit from a thinner vessel wall.

This sex-specific analysis may yield some clinical insights for subsequent research. First, the significantly elevated risk of DoCE linked to DCB in males—who constituted the majority of the study cohort—offers preliminary data that may support a preferential use of DES. Nevertheless, this finding requires confirmation in future prospective studies. Regarding female patients, who are frequently underrepresented, there is a need for studies with larger sample sizes or those specifically focused on females to elucidate the risk-benefit profile of DCB, especially in cases of small-vessel disease. In conclusion, these results underscore potential sex-based disparities in response to DCB and DES, which, if confirmed, could enhance the personalization of PCI decision-making in clinical settings.

Limitations

This analysis is subject to several limitations. First, the REC-CAGEFREE I study was only conducted in China with an East Asian population; therefore, extrapolation of these results to other ethnic groups requires caution. Second, our analysis has limitations inherent to subgroup analysis, including reduced statistical power for interaction testing and potential residual confounding. The observational design precludes definitive causal inferences, and the modest female representation underscores the need for larger, sex-balanced trials. Nevertheless, the consistency between our primary and subgroup findings strengthens the clinical validity of these observations. Third, because sirolimus-coated balloons (SCB) were not commercially available in China during the study period, patients were treated with paclitaxel-coated balloons only, which limits the extrapolation of our findings to patients treated with SCB. Fourth, given that the study relies on multi-level subgroup analyses, such repeated stratification poses challenges to statistical robustness, emphasizing the need for cautious interpretation of the results.

Conclusions

In males with de novo non-complex coronary disease, DCB was associated with a significantly higher 2-year DoCE risk than DES, whereas DCB-treated females exhibited a consistent yet non-significant trend toward higher DoCE risk. However, given the observational nature of the current study, the findings should be interpreted with caution and considered as hypothesis-generating only.

Abbreviations

DCB	Drug-coated balloon
DES	Drug-eluting stent
DoCE	Device-oriented composite endpoint
PoCE	Patient-oriented composite endpoint
NACE	Net adverse clinical events
TV-MI	Target vessel myocardial infarction
CPI-TLR	Clinically and physiologically indicated target lesion revascularization
CPI-TVR	Clinically and physiologically indicated target vessel revascularization
IPTW	Inverse probability of treatment weighting
SVD	Small vessel disease

Additional file: Tables S1-S4. Figures S1-S5. Table S1 Inclusion and exclusion criteria. Table S2 Testing of proportional hazard assumptions using Schoenfeld residuals. Table S3 The distribution of missing data. Table S4 Baseline characteristics of patients in the DCB group who received rescue DES. Fig. S1 Kaplan-Meier curves for the primary device-oriented composite endpoint and its components by sex at 2 years. Fig. S2 The incidence of DoCE in female and male patients allocated to the DCB group received rescue DES. Fig. S3 Kaplan-Meier curves for the secondary endpoints at 2 years. Fig. S4 Clinical outcomes in DCB/DES groups by sex at 2 years. Fig. S5 Clinical outcomes in female and male patients by DCB/DES at 2 years.

Declarations

Ethics approval and consent to participate

All participants included in the REC-CAGEFREE I provided informed consent for the data and samples to be used for scientific purposes. The study was conducted in accordance with the Declaration of Helsinki and was approved by the relevant Ethics Committees of all participating centers. The central ethics approval reference number is KY20202070-C-1.

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Authors’ contributions

LT obtained funding. LT and CG were responsible for the study concept and design. PW drafted the manuscript. JZ, HG, and TH participated in the pre-processing of the datasets. DS, QW, HJ, ZY, SW, YJ, HC, MY, LZ, SH, and ZL participated in the field investigation. GF and JL performed the statistical analysis, with ZJ, JX, and DW critically reviewing the analysis. XH and BZ provided input on critical thinking. SG, YO, and PWS critically reviewed and improved the drafts of the manuscript. All authors have read and approved the final manuscript.

Consent for publication

Not applicable.

Competing interests

PWS reports consulting fees from Sahajanand Medical Technologies, Novartis, and Merillife. SG reports consulting fees from Biosensors. ZJia is the founder of Beijing KeyTech Statistical Consulting and has stock in the company. All other authors declare no competing interests.

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Availability of data and materials

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

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Figure titles and legends

Table 1 Baseline patient characteristics

Fig. 1 Study flowchart

Fig. 2 Clinical outcomes by sex at 2 years

HR=hazard ratio, IPTW= inverse probability of treatment weighting.

Fig. 3 Clinical outcomes by sex and treatment assignment at 2 years

Fig. 4 Kaplan-Meier curves for the DoCE and its components at 2 years

Fig. 5 Forest plot of DoCE by sex and SVD/non-SVD at 2 years

SVD=small vessel disease.

Table 1 Baseline patient characteristics

	Female (N=698)	Male (N=1574)	P value Female vs. Male	Female		P value DCB vs. DES Female	Male		P value DCB vs. DES Male
				CB (N=364)	DES (N=334)		DCB (N=769)	DES (N=805)	
Demographics									
Age, years (SD)	64.6 (9.3)	59.9 (10.5)	<0.001	64.9 (9.3)	64.2 (9.3)	0.287	59.9 (10.3)	0.0 (10.7)	0.847
BMI, kg/m^2 (SD)	24.4 (3.5)	25.0 (3.4)	<0.001	24.2 (3.4)	24.6 (3.6)	0.133	25.0 (3.33)	25.0 (3.5)	0.921
Smoking (%)			<0.001			0.317			0.215
Current	20 (2.9)	744 (47.3)		13 (3.6)	7 (2.1)		347 (45.1)	397 (49.3)	
Former	1 (0.1)	247 (15.7)		1 (0.3)	0 (0.0)		122 (15.9)	125 (15.5)	
Never	677 (97.0)	583 (37.0)		350 (96.2)	327 (97.9)		300 (39.0)	283 (35.2)	
Comorbid conditions (%)									
Hypertension	69/698 (67.2)	97/1574 (57.0)	<0.001	40/364 (65.9)	229/334 (68.6)	0.510	22/769 (54.9)	475/805 (59.0)	0.109
Diabetes	18/698 (31.2)	02/1574 (25.5)	0.006	04/364 (28.6)	114/334 (34.1)	0.133	78/769 (23.1)	224/805 (27.8)	0.038
Insulin-treated diabetes	3/218 (24.3)	7/402 (21.6)	0.510	4/104 (23.1)	29/114 (25.4)	0.800	3/178 (24.2)	44/224 (19.6)	0.330
Hyperlipidemia	90/698 (84.5)	221/1574 (77.6)	<0.001	04/364 (83.5)	286/334 (85.6)	0.505	99/769 (77.9)	622/805 (77.3)	0.812
Left ventricular ejection fraction ≤40% or previous episode of heart failure	25/698 (3.6)	9/1574 (5.7)	0.047	13/364 (3.6)	2/334 (3.6)	>0.999	46/769 (6.0)	3/805 (5.3)	0.660
Left ventricular ejection fraction, %	61.3 (7.7)	58.9 (8.5)	<0.001	61.1 (7.9)	61.6 (7.4)	0.396	58.8 (8.6)	58.9 (8.4)	0.757
Previous stroke	58/698 (8.3)	31/1574 (8.3)	>0.999	32/364 (8.8)	6/334 (7.8)	0.731	62/769 (8.1)	9/805 (8.6)	0.784
Previous myocardial infarction	33/698 (4.7)	53/1574 (9.7)	<0.001	15/364 (4.1)	8/334 (5.4)	0.542	66/769 (8.6)	87/805 (10.8)	0.160
Previous PCI	62/698 (8.9)	17/1574 (13.8)	0.001	35/364 (9.6)	7/334 (8.1)	0.564	02/769 (13.3)	115/805 (14.3)	0.607
COPD	18/697 (2.6)	04/1567 (6.6)	<0.001	8/363 (2.2)	0/334 (3.0)	0.509	54/767 (7.0)	0/800 (6.3)	0.460
Chronic kidney disease[‡]	1/698 (10.2)	32/1567 (8.4)	0.195	7/364 (10.2)	34/334 (10.2)	>0.999	58/769 (7.5)	4/805 (9.2)	0.276
Peripheral vascular disease	21/698 (3.0)	9/1567 (5.0)	0.041	6/364 (1.6)	5/334 (4.5)	0.048	35/769 (4.6)	4/805 (5.5)	0.475
Clinical presentation (%)			<0.001			0.872			0.847
ST-elevation myocardial infarction	69 (9.9)	297 (18.9)		38 (10.4)	31 (9.3)		143 (18.6)	154 (19.1)	
Non-ST-elevation myocardial infarction	105 (15.0)	310 (19.7)		56 (15.4)	49 (14.7)		158 (20.5)	152 (18.9)	
Unstable angina	167 (23.9)	309 (19.6)		83 (22.8)	84 (25.1)		152 (19.8)	157 (19.5)	
Chronic coronary syndrome	357 (51.1)	658 (41.8)		187 (51.4)	170 (50.9)		316 (41.1)	342 (42.5)	

Table 1 Continued

	Female (N=698)	Male (N=1574)	P value Female vs. Male	Female		P value DCB vs. DES Female	Male		P value DCB vs. DES Male
				DCB (N=364)	DES (N=334)		DCB (N=769)	DES (N=805)	
PCI characteristics									
Number of lesions treated per patient (SD)	1.1 (0.3)	1.1 (0.3)	0.666	1.2 (0.4)	1.1 (0.3)	0.019	1.1 (0.3)	1.1 (0.3)	0.381
Number of DCB or DES per patient (SD)	1.3 (0.7)	1.3 (0.6)	0.123	1.4 (0.8)	1.3 (0.5)	0.017	1.3 (0.6)	1.3 (0.5)	0.022
Length of DCB or DES per patient, mm	31.6 (17.6)	31.2 (15.6)	0.572	33.1 (20.5)	30.0 (13.6)	0.021	32.4 (16.8)	30.0 (14.2)	0.003
IVUS or OCT guidance for PCI	69 (9.9)	166 (10.5)	0.687	30 (8.2)	39 (11.7)	0.164	78 (10.1)	88 (10.9)	0.669
Procedure time, min	39.8 (23.7)	40.5 (23.3)	0.507	39.1 (24.0)	40.4 (23.4)	0.484	39.5 (23.0)	41.4 (23.6)	0.101
Lesion characteristics (%)									
Total number	780	1749		417	363		849	900	
Lesion location			0.503			0.663			0.110
Left anterior descending artery	408 (52.3)	874 (50.0)		212 (50.8)	196 (54.0)		444 (52.3)	430 (47.8)	
Proximal left anterior descending artery	209 (26.8)	479 (27.4)		97 (23.3)	112 (30.9)		235 (27.7)	244 (27.1)	
Left circumflex artery	185 (23.7)	423 (24.2)		103 (24.7)	82 (22.6)		189 (22.3)	234 (26.0)	
Right coronary artery	187 (24.0)	452 (25.8)		102 (24.5)	85 (23.4)		216 (25.4)	236 (26.2)	
Pre-dilatation balloon types									
Semi-compliant balloon	615 (78.8)	1359 (77.7)	0.555	336 (80.6)	279 (76.9)	0.238	664 (78.2)	695 (77.2)	0.661
Non-compliant balloon	309 (39.6)	667 (38.1)	0.508	154 (36.9)	155 (42.7)	0.116	369 (43.5)	298 (33.1)	<0.001
Cutting or scoring balloon	467 (59.9)	1164 (66.6)	0.001	263 (63.1)	204 (56.2)	0.060	598 (70.4)	566 (62.9)	0.001
Small vessel (<3.0 mm)	433 (55.5)	792 (45.3)	<0.001	250 (60.0)	183 (50.4)	0.009	388 (45.7)	404 (44.9)	0.770
Bifurcation (%)	231 (29.6)	567 (32.4)	0.356	132 (31.7)	99 (27.3)	0.393	278 (32.7)	289 (32.1)	0.013
Thrombus aspiration	15 (1.9)	59 (3.4)	0.061	8 (1.9)	7 (1.9)	>0.999	29 (3.4)	30 (3.3)	>0.999
Diameter of DCB or DES per device, mm	2.94 (0.42)	3.06 (0.46)	<0.001	2.89 (0.42)	3.00 (0.42)	<0.001	3.04 (0.45)	3.08 (0.48)	0.044
Inflation duration of DCB or DES, s	35.1 (25.9)	33.9 (26.0)	0.256	56.0 (14.6)	9.6 (6.7)	<0.001	56.9 (13.5)	9.7 (6.8)	<0.001

Data are n (%), n/N (%). COPD=chronic obstructive pulmonary disease. DCB=drug-coated balloon. DES=drug-eluting stent. eGFR=estimated glomerular filtration rate. IVUS=intravascular ultrasound. OCT=optical coherence tomography. PCI=percutaneous coronary intervention. ‡Chronic kidney disease was defined as kidney damage (pathological abnormalities or markers of damage, including abnormalities in blood or urine tests or imaging studies) or an eGFR (by the Modification of Diet in Renal Disease formula) of <60 mL/min per 1.73 m² of body-surface area for at least 3 months.

Fig. 1 Study flowchart

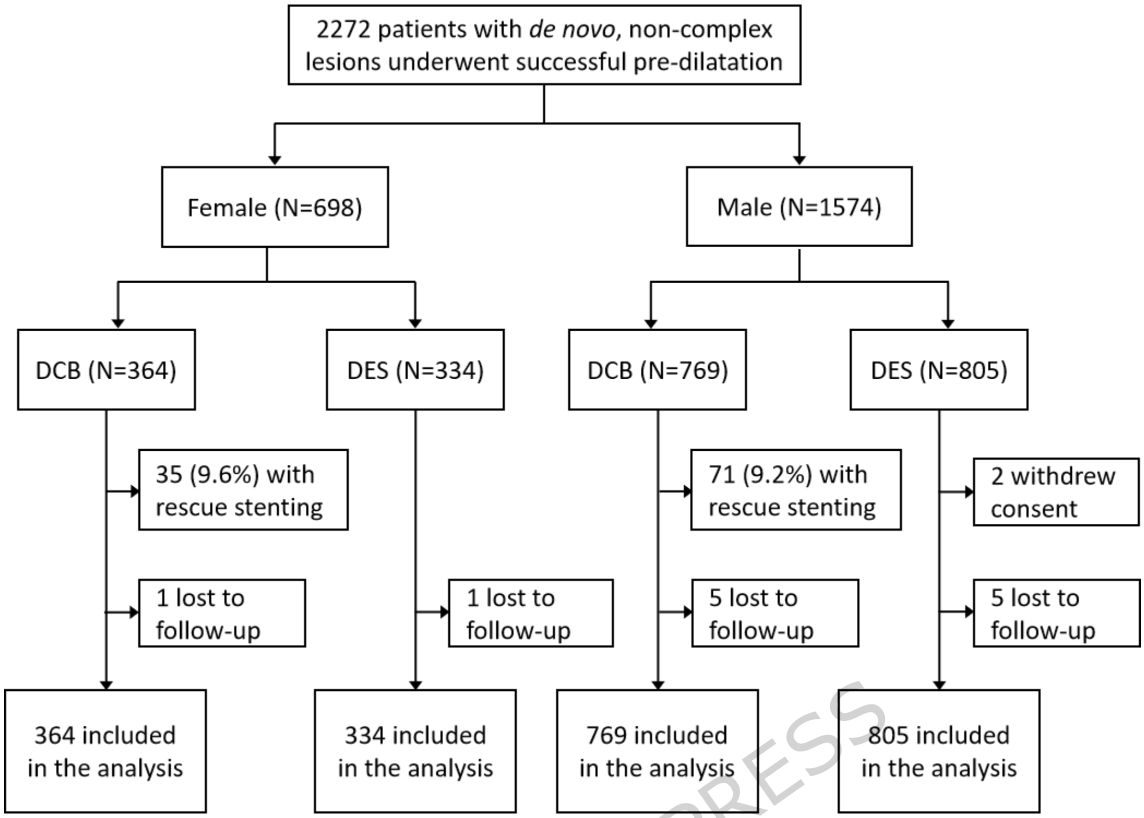
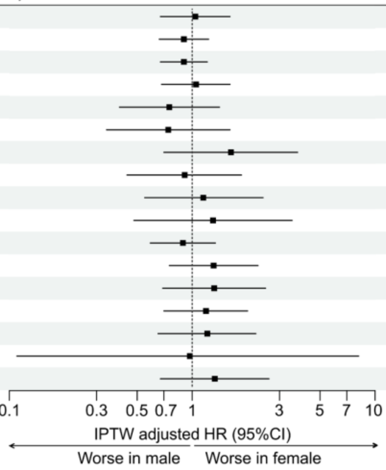


Fig. 2 Clinical outcomes by sex at 2 years

Outcomes	Female,n/N(%)	Male,n/N(%)	Crude HR (95%CI)	IPTW adjusted HR (95%CI)	Adjusted P value
Device-oriented composite endpoint	36/698 (5.2)	74/1574 (4.7)	1.10 (0.74−1.63)	1.04 (0.67−1.61)	0.877
Patient-oriented composite endpoint	67/698 (9.6)	157/1574 (10)	0.96 (0.72−1.28)	0.90 (0.66−1.23)	0.500
Net adverse clinical events	72/698 (10.3)	171/1574 (10.9)	0.95 (0.72−1.25)	0.90 (0.67−1.21)	0.487
Target vessel failure	38/698 (5.5)	77/1574 (4.9)	1.11 (0.75−1.64)	1.05 (0.68−1.61)	0.833
Any death	16/698 (2.3)	44/1574 (2.8)	0.82 (0.46−1.45)	0.75 (0.40−1.41)	0.375
Cardiovascular death	11/698 (1.6)	29/1574 (1.9)	0.85 (0.43−1.70)	0.74 (0.34−1.61)	0.453
Any stroke	11/698 (1.6)	14/1574 (0.9)	1.77 (0.80−3.89)	1.63 (0.70−3.77)	0.254
Any myocardial infarction	13/698 (1.9)	33/1574 (2.1)	0.89 (0.47−1.69)	0.91 (0.44−1.86)	0.789
Target−vessel myocardial infarction	13/698 (1.9)	26/1574 (1.7)	1.13 (0.58−2.19)	1.15 (0.55−2.44)	0.709
Periprocedural myocardial infarction	8/698 (1.1)	11/1574 (0.7)	1.64 (0.66−4.08)	1.30 (0.48−3.52)	0.609
Any revascularization	38/698 (5.5)	92/1574 (5.9)	0.93 (0.63−1.35)	0.89 (0.59−1.34)	0.578
Target lesion revascularisation	22/698 (3.2)	42/1574 (2.7)	1.18 (0.70−1.97)	1.31 (0.75−2.29)	0.350
Clinically and physiologically indicated	17/698 (2.5)	32/1574 (2.1)	1.19 (0.66−2.15)	1.32 (0.69−2.52)	0.397
Target vessel revascularisation	25/698 (3.6)	49/1574 (3.2)	1.15 (0.71−1.86)	1.19 (0.70−2.01)	0.523
Clinically and physiologically indicated	19/698 (2.8)	36/1574 (2.3)	1.19 (0.68−2.07)	1.21 (0.65−2.23)	0.548
Definite or probable device or vessel thrombosis	1/698 (0.1)	6/1574 (0.4)	0.38 (0.05−3.12)	0.97 (0.11−8.13)	0.974
BARC type 3 or 5 bleeding	14/698 (2.0)	29/1574 (1.9)	1.08 (0.57−2.05)	1.33 (0.67−2.63)	0.415



HR=hazard ratio, IPTW= inverse probability of treatment weighting.

Fig. 3 Clinical outcomes by sex and treatment assignment at 2 years

Outcomes at 2 years	DCB, n/N (%)	DES, n/N (%)	Crude HR (95%CI)	IPTW adjusted HR (95%CI)	Adjusted P value	P for interaction
Device−oriented composite endpoint						
Female	23/364 (6.3)	13/334 (3.9)	1.64 (0.83−3.24)	1.55 (0.78−3.11)	0.210	0.575
Male	49/769 (6.4)	25/805 (3.1)	2.08 (1.29−3.37)	2.28 (1.40−3.70)	0.001	
Patient−oriented composite endpoint						
Female	44/364 (12.1)	23/334 (6.9)	1.80 (1.09−2.98)	1.80 (1.08−3.01)	0.024	0.450
Male	90/769 (11.7)	67/805 (8.4)	1.43 (1.04−1.96)	1.52 (1.10−2.09)	0.011	
Net adverse clinical events						
Female	45/364 (12.4)	27/334 (8.1)	1.56 (0.97−2.51)	1.58 (0.97−2.57)	0.060	0.460
Male	93/769 (12.1)	78/805 (9.7)	1.26 (0.93−1.70)	1.35 (1.00−1.84)	0.050	
Target vessel failure						
Female	25/364 (6.9)	13/334 (3.9)	1.79 (0.91−3.49)	1.72 (0.87−3.42)	0.120	0.926
Male	49/769 (6.4)	28/805 (3.5)	1.86 (1.17−2.95)	2.05 (1.28−3.27)	0.003	
Any death						
Female	10/364 (2.8)	6/334 (1.8)	1.53 (0.56−4.22)	1.26 (0.45−3.54)	0.660	0.885
Male	27/769 (3.5)	17/805 (2.1)	1.67 (0.91−3.07)	1.75 (0.95−3.22)	0.070	
Cardiovascular death						
Female	6/364 (1.7)	5/334 (1.5)	1.10 (0.34−3.62)	0.86 (0.26−2.90)	0.810	0.301
Male	20/769 (2.6)	9/805 (1.1)	2.34 (1.07−5.14)	2.49 (1.13−5.50)	0.024	
Any stroke						
Female	7/364 (1.9)	4/334 (1.2)	1.61 (0.47−5.51)	1.60 (0.46−5.51)	0.460	0.870
Male	8/769 (1.1)	6/805 (0.8)	1.41 (0.49−4.05)	1.51 (0.52−4.37)	0.450	
Any myocardial infarction						
Female	8/364 (2.2)	5/334 (1.5)	1.47 (0.48−4.50)	1.54 (0.49−4.80)	0.460	0.440
Male	15/769 (2.0)	18/805 (2.3)	0.88 (0.44−1.74)	0.98 (0.49−1.96)	0.950	
Target−vessel myocardial infarction						
Female	8/364 (2.2)	5/334 (1.5)	1.47 (0.48−4.50)	1.54 (0.49−4.80)	0.460	0.630
Male	13/769 (1.7)	13/805 (1.6)	1.05 (0.49−2.27)	1.12 (0.52−2.42)	0.780	
Periprocedural myocardial infarction						
Female	4/364 (1.1)	4/334 (1.2)	0.96 (0.23−3.67)	0.83 (0.21−3.35)	0.800	0.730
Male	6/769 (0.8)	5/805 (0.6)	1.26 (0.38−4.13)	1.56 (0.47−5.15)	0.460	
Any revascularization						
Female	26/364 (7.2)	12/334 (3.6)	2.03 (1.02−4.02)	2.23 (1.12−4.45)	0.023	0.350
Male	52/769 (6.9)	40/805 (5.0)	1.38 (0.92−2.09)	1.46 (0.96−2.22)	0.080	
Target lesion revascularization						
Female	17/364 (4.7)	5/334 (1.5)	3.17 (1.17−8.59)	3.37 (1.23−9.20)	0.018	0.899
Male	32/769 (4.2)	10/805 (1.3)	3.43 (1.68−6.97)	3.96 (1.93−8.13)	<.001	
Clinically and physiologically indicated						
Female	12/364 (3.3)	5/334 (1.5)	2.22 (0.78−6.31)	2.40 (0.84−6.88)	0.100	0.759
Male	23/769 (3.0)	9/805 (1.1)	2.72 (1.26−5.89)	3.09 (1.42−6.75)	0.005	
Target vessel revascularization						
Female	20/364 (5.6)	5/334 (1.5)	3.74 (1.40−9.97)	4.07 (1.52−10.93)	0.005	0.461
Male	34/769 (4.5)	15/805 (1.9)	2.42 (1.32−4.44)	2.77 (1.50−5.12)	0.001	
Clinically and physiologically indicated						
Female	14/364 (3.9)	5/334 (1.5)	2.60 (0.94−7.21)	2.87 (1.03−8.05)	0.044	0.606
Male	23/769 (3.0)	13/805 (1.6)	1.88 (0.95−3.71)	2.12 (1.06−4.22)	0.033	
Definite or probable device or vessel thrombosis						
Female	0/364 (0.0)	1/334 (0.3)	0 (0−Inf)		>0.999	>0.999
Male	4/769 (0.5)	2/805 (0.2)	2.10 (0.39−11.48)	2.49 (0.45−13.87)	0.300	
BARC type 3 or 5 bleeding						
Female	7/364 (1.9)	7/334 (2.1)	0.92 (0.32−2.62)	0.87 (0.30−2.52)	0.800	0.317
Male	9/769 (1.2)	20/805 (2.5)	0.47 (0.21−1.03)	0.53 (0.24−1.16)	0.110	

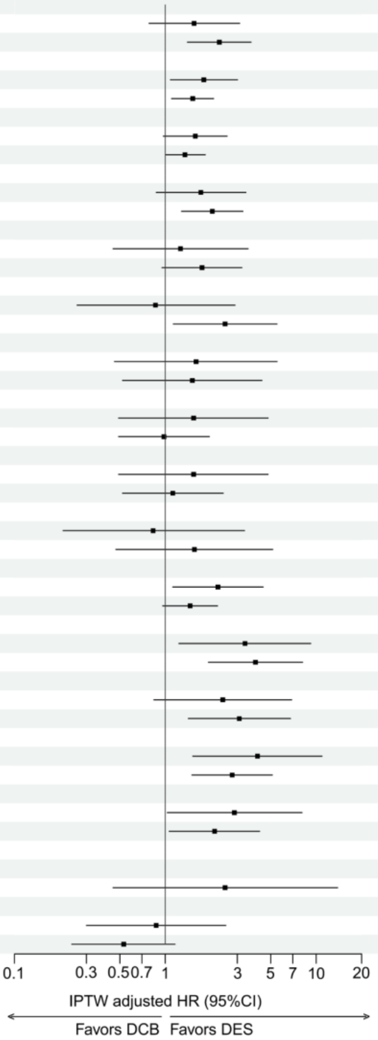


Fig. 4 Kaplan-Meier curves for the DoCE and its components at 2 years

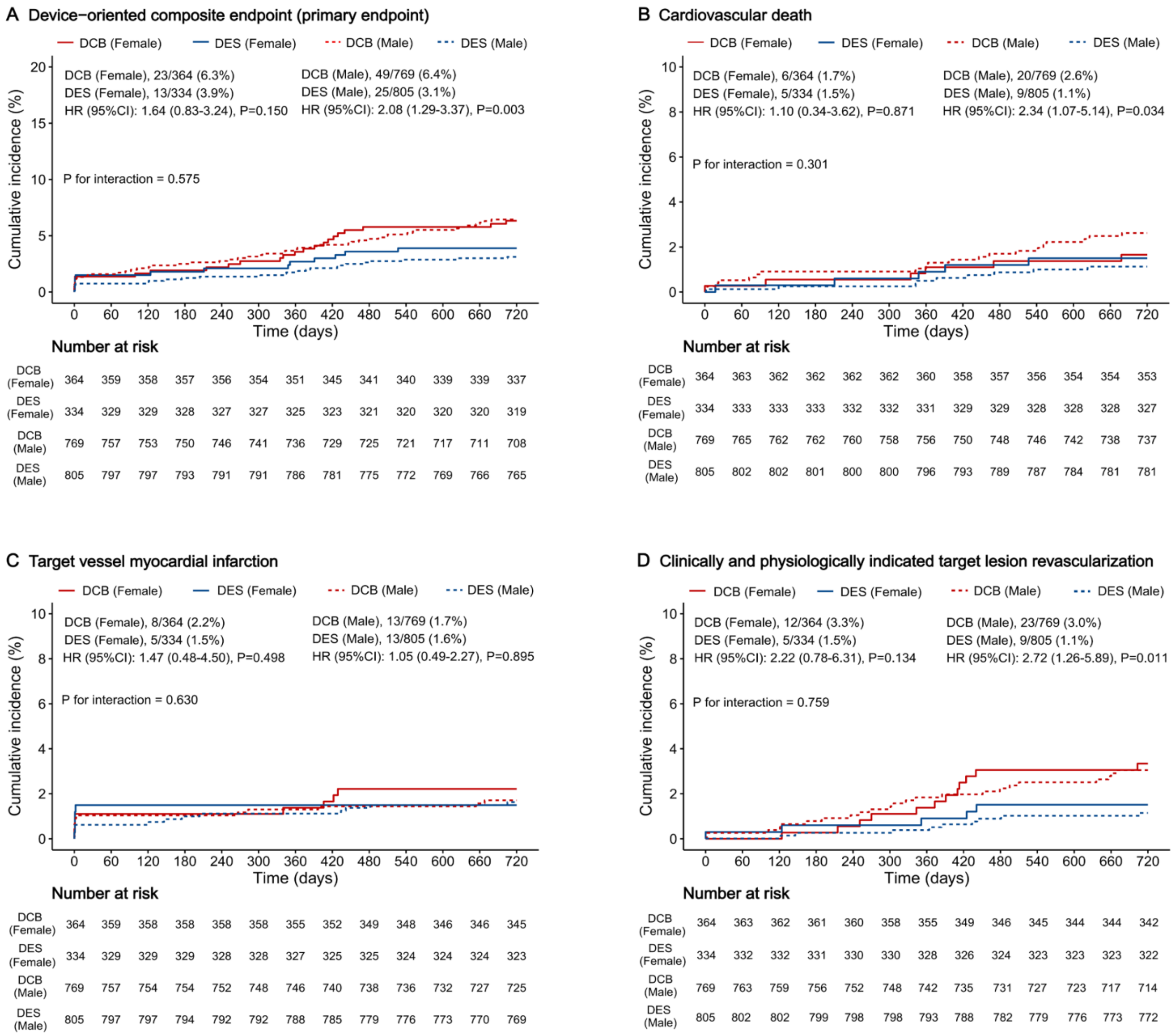


Fig. 5 Forest plot of DoCE by sex and SVD/non-SVD at 2 years

	DCB,No(%)	DES,No(%)	Crude HR (95%CI)	IPTW adjusted HR (95%CI)	Adjusted P value	P for interaction
Female	23/364 (6.3%)	13/334 (3.9%)	1.64 (0.83−3.24)	1.55 (0.78−3.11)	0.210	
SVD	8/201 (4.0%)	10/163 (6.1%)	0.65 (0.26−1.64)	0.69 (0.26−1.85)	0.465	0.007
Non-SVD	15/163 (9.2%)	3/171 (1.8%)	5.41 (1.57−18.7)	5.42 (1.53−19.24)	0.009	
Male	49/769 (6.4%)	25/805 (3.1%)	2.08 (1.29−3.37)	2.28 (1.40−3.70)	0.001	
SVD	19/329 (5.8%)	12/339 (3.6%)	1.65 (0.80−3.40)	1.80 (0.87−3.76)	0.115	0.408
Non-SVD	30/440 (6.9%)	13/466 (2.8%)	2.49 (1.30−4.77)	2.70 (1.40−5.23)	0.003